



Chapter 5: Contract Management

5.1 Method of Work Execution

➤ Survey

- Desk Study (completed)
- Preliminary Survey (completed)
- Detailed Survey (going on)

➤ Design, Drawing and Specification

➤ Cost Estimate

➤ Approval

➤ Tendering Process

- Tender Notice
- Tender document preparation
- Buying/ selling tender documents by prospective bidder
- Submission of bid/tender by bidders
- Evaluation of bid
- Selection of contractor
- Signing / award of contract

Method of Work Execution

Generally owner have following choices for work execution

1) Amanat:

- Works are executed by the owner itself, by hiring workers and supplying materials and equipment.
- The record of labor employed, materials and equipment are maintained in book keeping.
- It is useful for small work, if it is necessary to do immediately and expertise are available with in the organization.
- It saves the cost and keeps the staff busy in work.

2) Negotiation:

- Contract is awarded through negotiation with the contractor.

3) Contract:

- It is the agreement between two or more parties to do or not to do any work.
- It is legal and time bound
- Offer and acceptance must be there.

Method of Work Execution

As per **public procurement act of Nepal**, there are following provisions of work Execution.

A) Sealed Competitive Bidding: Contractor is selected based on the competition among the bidders (contractors).

1) International Competitive Bidding (ICB)

- It will be as per agreement with donor agency.
- If Nepalese contractors have no experience or not capable.
- If no bid is submitted by Nepalese contractors in stated time.
- Notice on invitation to tender and all tender or prequalification documents shall be published in English language.
- In publishing a notice a period of at least 45 days shall be given
- Notice shall be published in daily newspaper of national and international communication media.
- Notice shall be placed in website of concerned entity or that of the Public Procurement Monitoring office (PPMO).

Method of Work Execution

2) National / Local Competitive Bidding (NCB/LCB)

- Contractors are invited from within the nation or local.
- If Construction work estimated exceeds Rs. Twenty lakhs.
- Good & services estimated exceeds Rs. Ten lakhs.
- In publishing a notice a period of at least 30 days shall be given
- Notice shall be published in daily newspaper of national and it may also be published in international communication media.
- Notice shall be placed in website of concerned entity or that of (PPMO).

B) Sealed Quotation (SQ)

- Time required for this is shorter than NCB.
- If Construction work estimated less or equal to Rs. Twenty lakhs.
- Good & services estimated less or equal to Rs. Ten lakhs.

Method of Work Execution

C) Direct Procurement (DP)

- Construction work up to Rs. Five lakhs
- Goods or Consultancy services or other services up to Rs. Three lakhs

D) User's Committee (UC)

- Works up to Rs. Sixty lakhs can do by User's Committee.
- This Method is used for executing small labor intensive works where the objective is to:
 - Provide employment and income directly to persons living in the project area
 - Increase the utilization of local know how, appropriate technologies and materials
 - Handover the completed works for operation and maintenance directly by persons living in project area.

E) Procurement Under Special Circumstances

- Emergency/ Special circumstances
- Depends on contract amount

Types of Contract

There are two broad categories of construction contracts

A) Price Based

- Lump sum or unit rate contracts
- Price or rates are submitted by contractor in tender

B) Cost Based

- Cost-reimbursable and target cost contract
- Actual cost incurred by the contractors is reimbursed together with a fee for overheads and profit.

Lump Sum Contract:

- Some times called drawings & specifications contract
- If the quantities of work are difficult to measure, it may be useful e.g. ground water pipe line installation.
- It shall specify that the construction entrepreneur shall be responsible for all types of risks and liabilities associated with the construction work.
- If the financial liabilities increases as a consequence of an order issued by the public entity that involves any change in the construction work upon making the contract, the public entity shall bear the liabilities.

Lump Sum Contract

Advantages:

- The final price is known by the owner before the work commences.
- The contractor has more incentive to reduce cost & to increase profit.
- The contractor hopes to complete the job as quickly as possible to minimize overhead, to maximize profit and to move to next job.

Disadvantages:

- Changes in drawings and specifications can be very expensive and source of trouble (i.e. contract has very limited flexibility for design changes).
- The contractor carries much of the risk so, the tendered price may include high risk contingency.
- Competent contractor may decide not to bid to avoid high risk.

Unit price (BOQ) Contract

- It may be entered where the quantity of a construction work is difficult to be ascertained.
- where the construction work is to be procured on the basis of unit price set forth in the bill of quantity.
- The bidder has to include in such unit price the materials, labor and other matters required to complete the proposed construction work.
- The public entity shall make such payment on the basis of the unit of work actually done and measured in the field.

Advantages:

- Fair basis for competition
- In comparing with lump-sum contract,
 - Changes in contract documents can be made easily by the owner
 - Lower risk for contractor

Disadvantages:

- The exact final price of the project is not known until the completion of the project.
- Up to date account and record keeping is necessary.
- Possibility of submitting unbalanced bid.

Cost Reimbursable Contract

- This contract may be entered into for procuring a construction work involving high risk and unpredictable condition when a construction entrepreneur refuse to, or be unable to, perform the work under a unit price contract.
- While making the payment, it includes the actual cost incurred by entrepreneur, overheads and profit set forth in the approved cost estimate.

Advantages:

- Start construction without waiting the whole set of drawings and specifications.
- More flexibility for the owner to make changes as work progresses.
- Draw the contractor expertise during design.

Disadvantages:

- It is difficult to predict the final cost and the distribution of it which may cause financial problems to the owner.
- Contractor pays less attention to cost control

Design and Built Construction (Turn Key Contract)

- A design and built contract may be entered for procuring the **design and built of any construction** from the same entrepreneur.
- The work shall commence only after the public entity has thorough its technician or group of technicians **examined and approved the design** of construction work.
- In this contract, the owner enters into a single agreement by which the design and build **contractor agrees to perform both the design and construction** of the project.
- As well as being responsible for faulty workmanship in construction, the contractor is also **liable for any deficiencies in design under this**.
- It is usually the preferred contracting method **under tight schedule circumstances and it is intended to save time**.

Design and Built Construction (Turn Key Contract)

Advantages:

Disputes between owner and contractor which arises because of design fault or construction (quality) fault, such situation is reduced in this method of contract.

The contractor is unable to look to the owner for additional compensation.

In this contract, the project can often be completed within a shorter period of time than with the traditional three-party arrangement.

Since a design built project can be designed and constructed in phases, the contractor is able to order necessary materials for subsequent phases ahead of time perhaps at a reduced cost.

Contractor's control over design details allows the contractor to use familiar construction methods and processes in building the structure, with the result of efficient construction, saving ultimately benefit the owner.

Design and Built Construction (Turn Key Contract)

Disadvantages:

- For owner it is often difficult to effectively compare the various preliminary design proposals. The designs will probably not be uniform.
- The owner's input on the detailed design of the structure will be limited.
- The owner may not obtain the lowest cost for the project
- There is only limited scope for the owner to make changes to his requirements.
- The owner has less control and influence over design matters due to which quality may be impaired.

Target Contract:

- In this contract, a target of cost is fixed before execution.
- If the work is completed as per target, target amount and agreed percentage is paid.

Construction Arrangement

a) Traditional Approach:

- In this approach, there is cooperation between **owner and consultant**
- There is also cooperation between **owner and contractor/subcontractors** as well
- But **no cooperation** between **consultant and contractor**.
- Because of this, there may arise problem in smooth execution of the project and may impact in cost and quality of work. So, it may create conflict among the three stakeholders (i.e. owner, Consultant and contractor)

b) Design-Built Approach:

- Contractor design and built the facility
- No role of consultant
- Fast and estimate high

c) BOOT Model:

- BOOT- Build Own Operate & Transfer
- Build and own by private organization, operate up to specified period and transfer to the government with free of cost.

Construction Arrangement

d) Owner-Builder Approach:

- Owner builds the facility using it's own resources.
- It has it's own design section /construction department.

e) Construction Management Approach:

A team of construction professionals are selected to co-ordinate each activity.

f) Labor Contract

g) Material Contract

h) Equipment Contract

5.3 Tendering Process

Tender, Bid, Bid Price, Ask Price, Tendering and Tender Document:

- A **tender** is an invitation to bid for a project or accept a formal offer such as a takeover bid.
- A **tender** is essentially an official letter that the organizations have to send for the procurement of the goods and services required on a large scale.
- For the execution of projects also **tender notice** is required. Thus, the **tenders** are filled when an invitation is received for bidding.
- In business terms, **Tender** is an invitation from a buyer who requires particular goods or services to suppliers of those products to put in negotiable price bids.
- A **bid price** is the highest price that a buyer is willing to pay for a goods. It is usually referred to simply as the "bid". In bid and ask, the bid price stands in contrast to the ask price or "offer", and the difference between the two is called the bid.
- **Tendering** usually refers to the **process** whereby governments and financial institutions invite bids for large projects that must be submitted within a finite deadline.
- **Tendering** is the process of making an offer, bid or proposal, or expressing interest in response to an invitation or request for tender.
- A **tender document** is a written request sent to potential suppliers to ask for information required for the buyer.

Tendering Process

Tender Notice:

- Invitation by client to eligible bidders / contractors to submit proposal (bid / tender) to construct the facility or to supply materials through public notice (media).

Process to be completed before inviting tender:

- Desk Study
- Field survey
- Design, drawing, specification
- Rate analysis & cost estimation
- Approval of budget
- Tender / bid documents preparation
- Tender notice publication

Tendering Process

Essential for Tender Notice:

- Implementing agency (Client's name / address)
- Notice No
- First date of publication
- Contract No, description of work and location
- Bid Security / Earnest money deposit
- Bidding documents (details of job, including items of work, drawings, specification and location etc.) fee.
- Office to buy bidding documents and submit tender
- Last date of submission of bid
- Cost estimate of the project
- Name of authority to accept or reject the tender
- Date, place and time for opening of the tender

Tendering Process

Tender Document: It is a document prepared by the concerned firm making invitation to bid for submission by bidder by filling up the price or rate. It includes the following documents.

- Bid form
- Sample (Format) of Letter of Acceptance (LOA), General Condition of Contract (GCOC) and Special Condition of Contract (SCOC)
- Specifications
- BOQ
- Tender drawings
- Sample form of bid security, performance security, advance guaranty
- Qualification certificates

While preparing tender document, following standard bidding documents (SBD) are used.

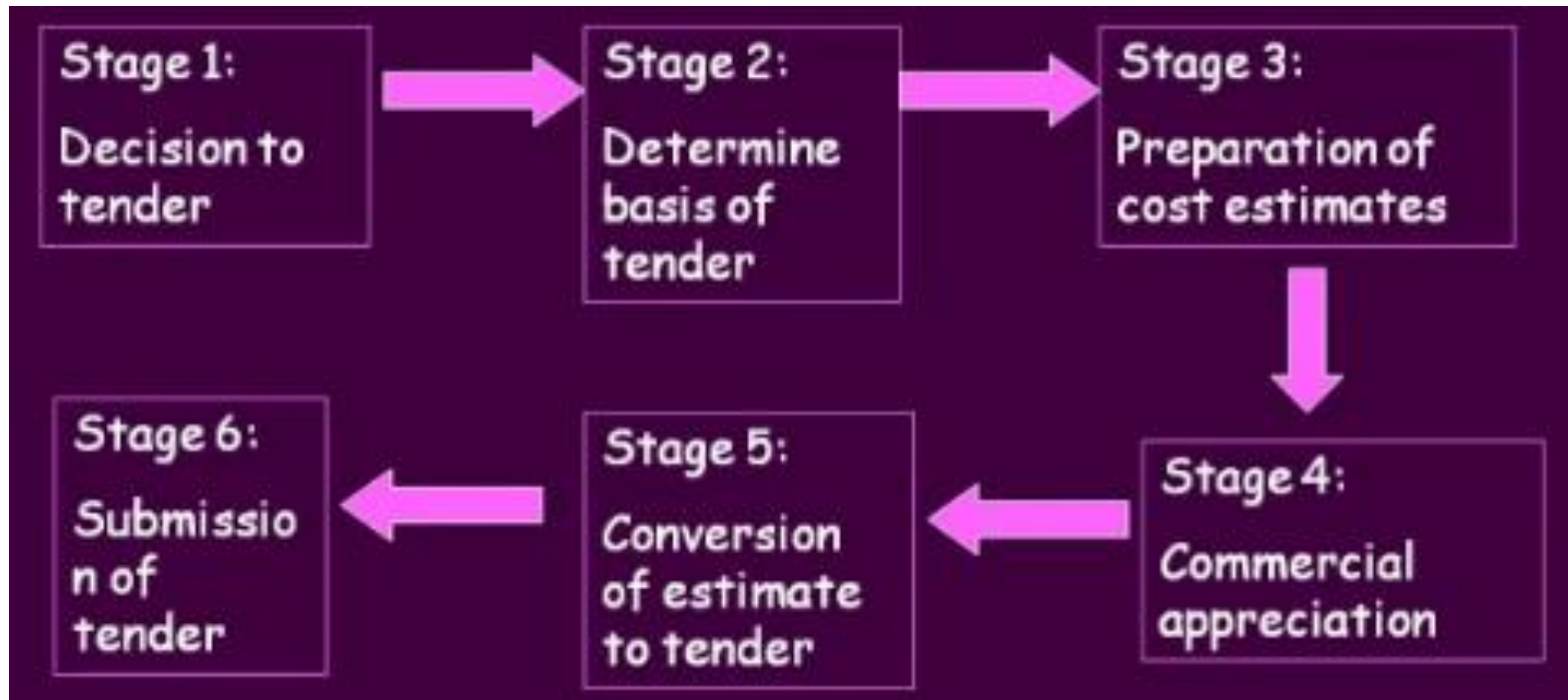
FIDIC: Federation Internationale Des Ingenieurs Conseils. (Which means “International Federation of Consulting Engineers”)

WB: World Bank

ADB: Asian Development Bank

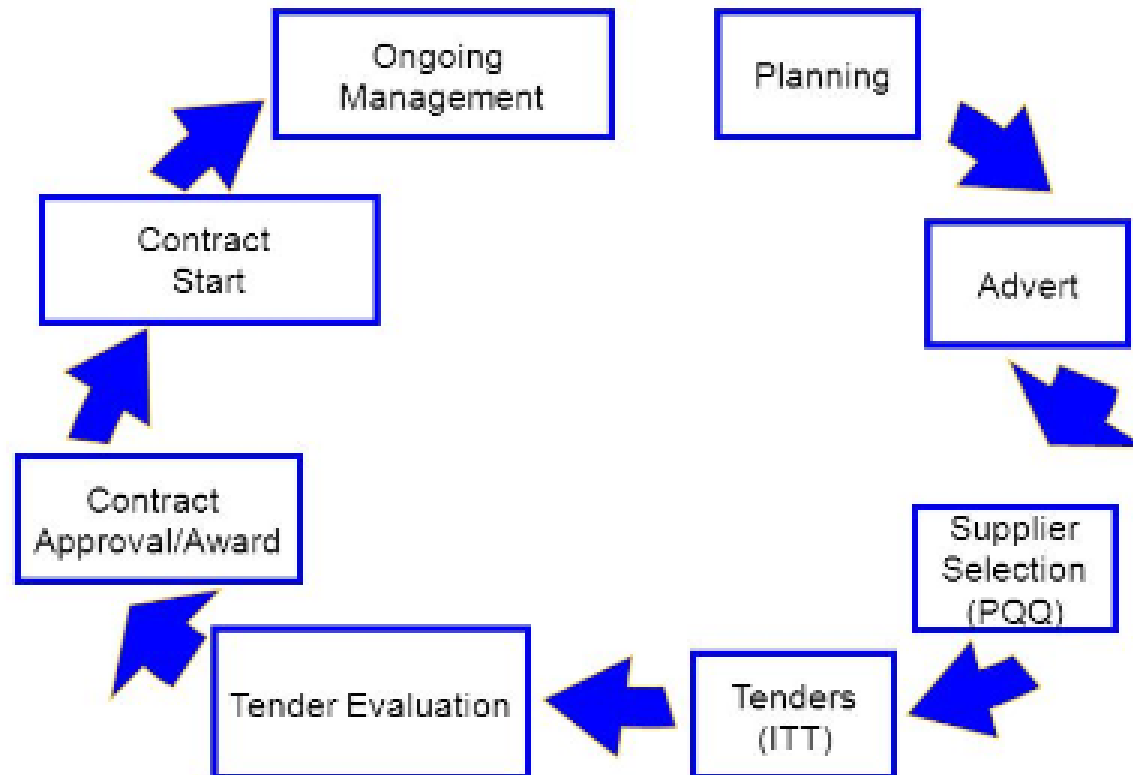
Tendering Process

General Tendering Process



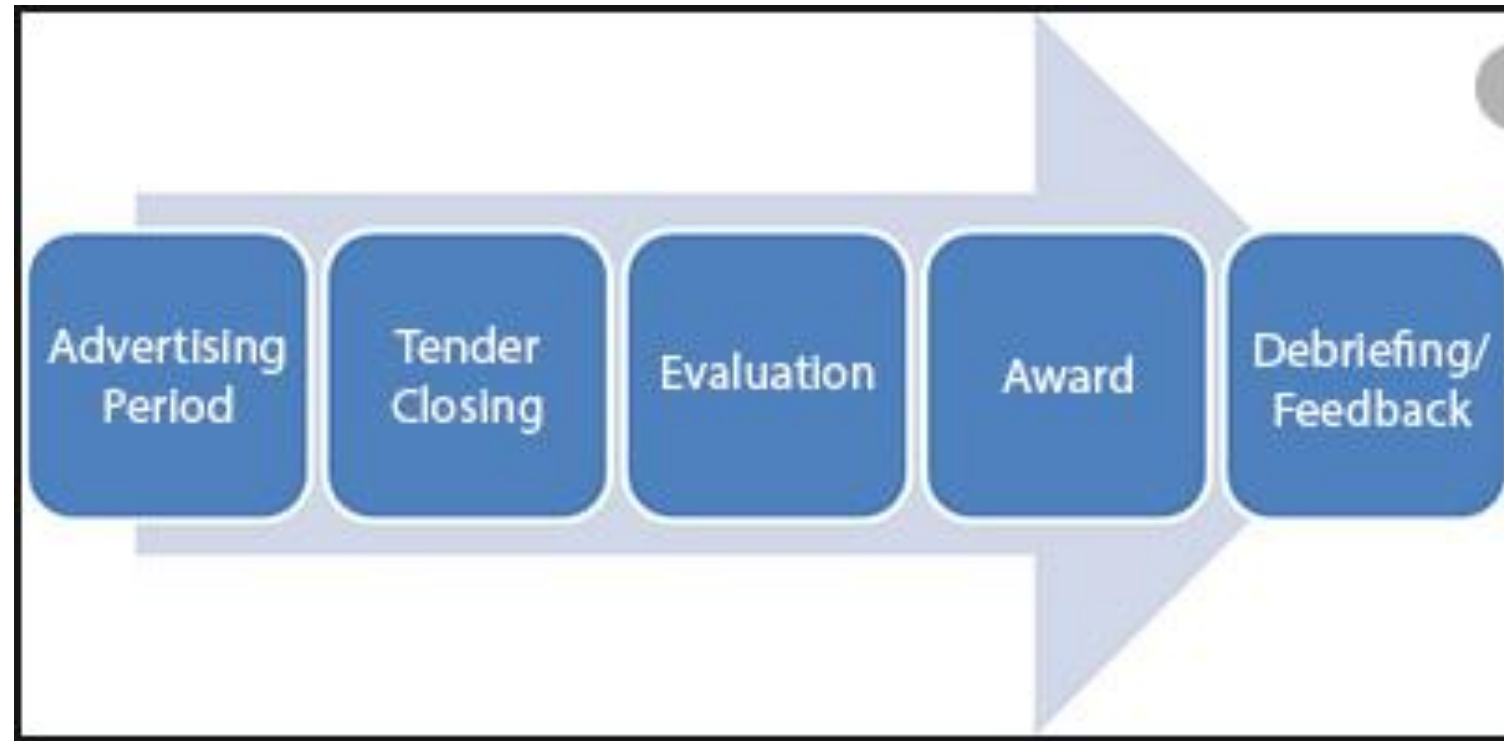
Tendering Process

General Tendering Process



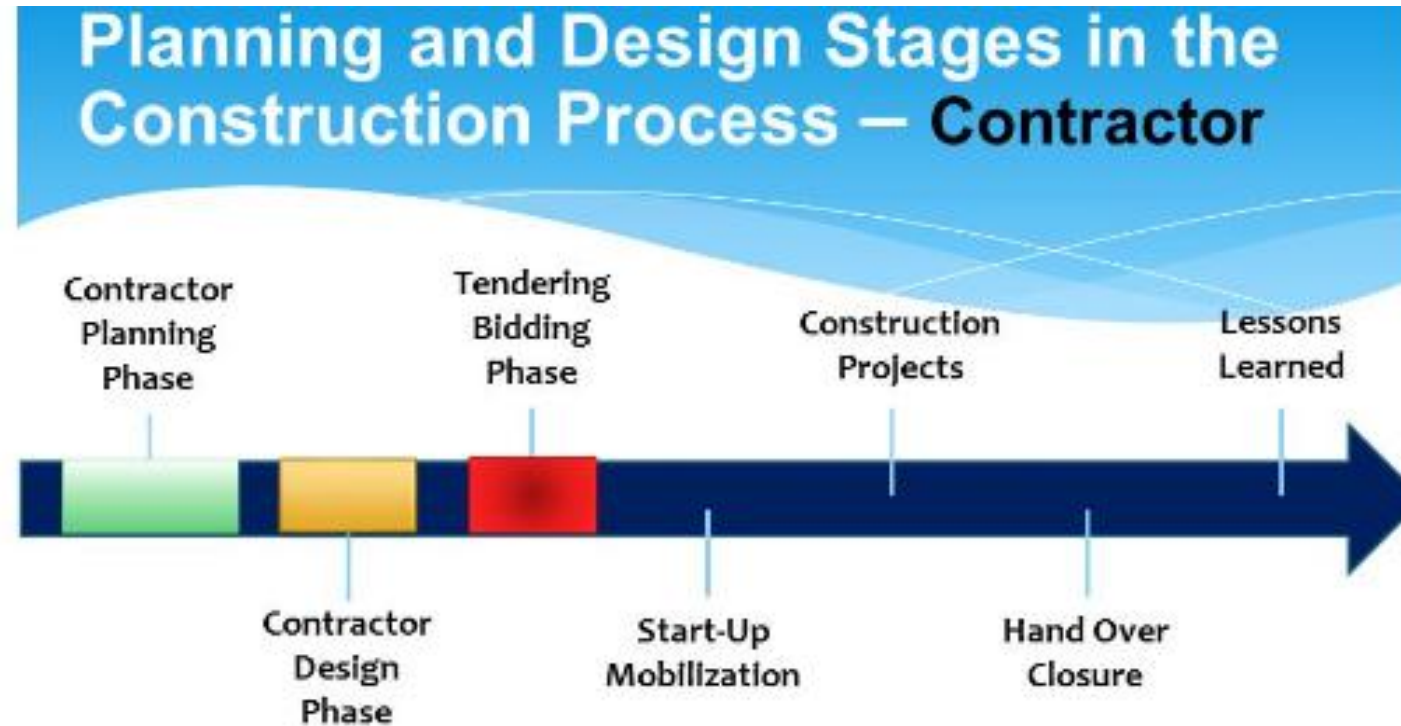
Tendering Process

Tendering Process



Tendering Process

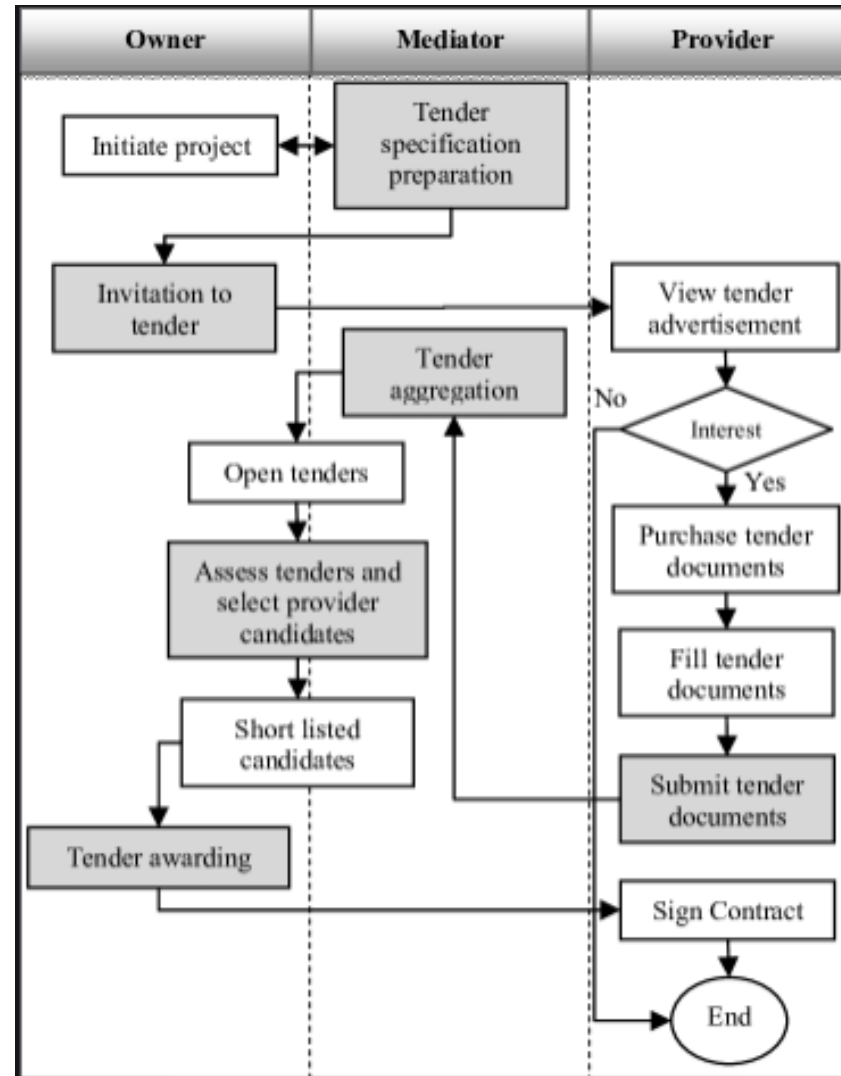
Construction Tender Process



Each stage has a systematic set of interfaced proceedings, events, activities and processes.

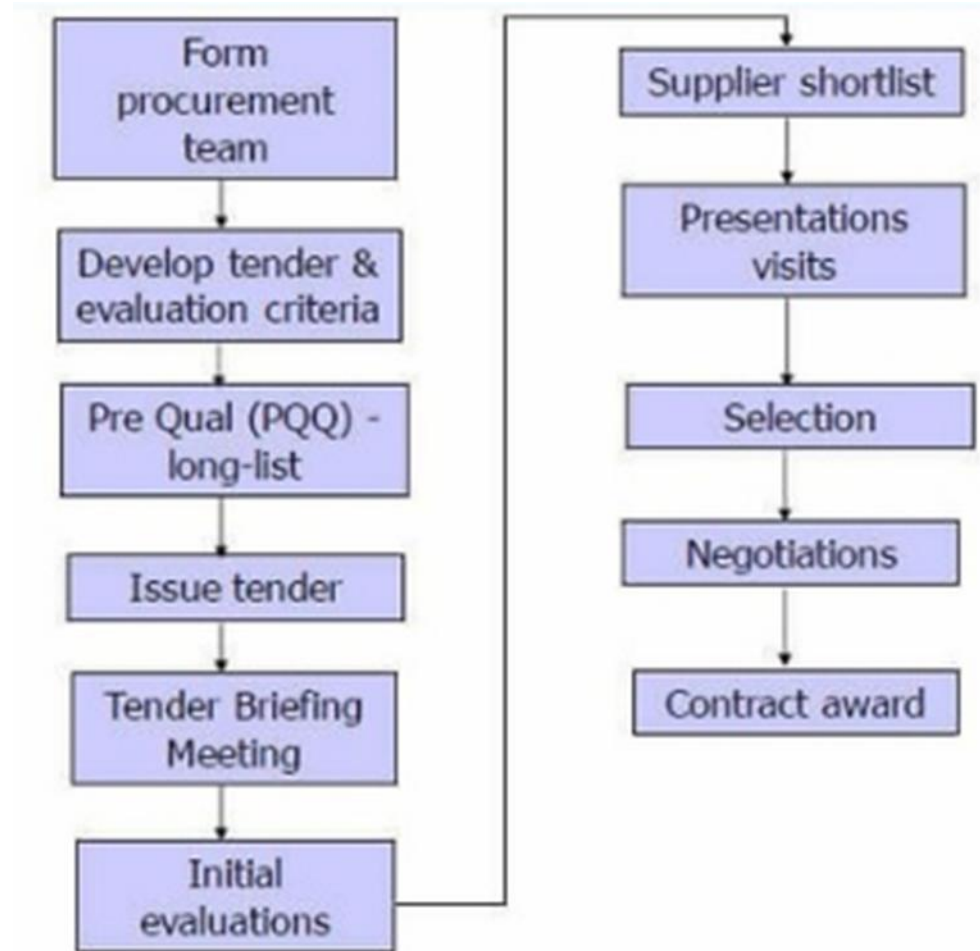
Tendering Process

General Tendering Process



Tendering Process

Tender Procurement Process



Tendering Process

Steps in a Typical E-tendering Process

Pre-qualification & registration	Pre-Qualification Registration
	Issue User Name and Password
Public invitation	Tender Advertisement
	Tenderer Views Tender Advertisement and Notice
Tender submission	Tenderer Registration to Tender for a Project
	Download Tender Document
	Addenda Distributed by Principal
	Tenderer Submits Tender
Close of Tender	Close Tender
	Principal Opens Tender
Tender Evaluation	Tender Evaluation Process
	Request for Information
Award of Tender	Award Tender/Acceptance of Tender
	Sign the Formal Agreement
Archiving	Retention of Document

Tendering Process

Condition of Contract:

- Contracts are legal agreement binding both the owner and the contractor.
- It is very important to have all the terms clearly spelled out for easy functioning of the contract & to minimize the disputes between the contracting parties.
- Such conditions are called Condition of Contract (COC). There are two types of COC.
 - a) General Condition of Contract (GCOC):** It is almost same for all types of contract. The documents are copied from FIDIC's / WB's / ADB's standard bidding document.
- Definition and interpretations
- Time for completion and delays
- Payment to contractors
- Execution of work and measurement of completed work
- Breach of contract
- Arbitration- settlement of contract
- Suspension of work
- General obligation of contractors
- Labor and labor welfare

Tendering Process

b) Special Conditions of Contract (SCOC)

Definition:

- It is special for specific project / contract.
- Some conditions may be changed by mutual agreement between the parties.
- Which may be based on law of country, nature of work etc.
- All the clauses of GCOC can't be applicable universally. So, they need modification, addition or alternation depending up on the type and nature of the project.
- Some clauses may need revisions because of legal requirement of the countries.
- Some country specific/ project specific/ location specific additional clauses might also be required. These are provided in separate part and are called Special Condition of Contract.

Tendering Process

Following aspect should therefore be covered in SCOC.

- **Detailing** of provisions which have been treated generally in GCOC.
- **Deletion** of any provisions which have been treated generally in GCOC.
- **Amendment** of any provision in GCOC to cover more specifically or appropriately the requirements of contract in hand.
- **Addition** of further provision deemed necessary for the satisfactory administration and execution of the contract.

Prequalification

Prequalification (PQ):

- The assessment by an implementing agency of **suitability of firms to carry out a particular contract** prior to being invited to submit a bid is a process called prequalification.
- A **process of evaluating and determining** whether potential bidders have the skill, judgment, integrity, sufficient financial resources, and ability necessary to the faithful performance of a contract for construction or repair work.
- It is a kind of short listing of eligible bidders to avoid crowding of them.
- It **ensures** that the invitation to bids extends only to those perspective bidders who have **adequate capacity and resources** to perform the particular contract.
- Prequalification requirement shall be basically for the **large and complex works**, industrial plant with **high costs goods procurement** or to identify the potential bidders.
- In order to prequalify for the proposed works, the **applicant should meet all the criteria** as defined in the Prequalification document.

Prequalification

Prequalification Process:

- Each prospective bidder on contracts identified for prequalification shall submit an application on the approved prequalification application form in order to become prequalified.
- The approved prequalification application form will require information to be provided on the ownership of the firm, experience of firm's personnel, any affiliations with other construction firms, bonding capacity, financial resources, the type of work performed by the firm, and other objective criteria rationally related to the particular contract to be awarded.
- The Bid Agent shall ensure that applications prepared for the projects comply with this policy and State law.
- The application is to be submitted by the deadline established in the notice of prequalification.
- The notice must be posted on a national news paper giving period of minimum of two weeks.
- The act of submitting the application does not permit the firm to submit a bid.
- Incomplete applications will be rejected.

Prequalification

The application shall at a minimum, address the following items:

a) Organizational Structure

- The firm shall provide a list of all owners, officers, partners, or individuals authorized to represent or conduct business for or sign legal documents for the firm.
- This list must include the full legal name, typed or printed in a clear legible form.
- Firms experiencing changes in ownership, organizational structure, or material changes in assets must inform the Bid Agent prior to the award of a contract.
- Failure of the firm to comply with this requirement may result in the termination of any contract awarded

b) Classification

- The firm shall indicate the type(s) of work the firm's workforce and equipment normally perform, licensure, and other pertinent information.

c) Experience

- The firm shall furnish a history documenting experience with projects of similar size, scope, or complexity involving the type(s) of work for which prequalification is requested.

d) Financial

- Firms will be required to provide a minimum of one year of complete audited financial statements.

Prequalification

e) Litigation / Claims

- The firm will be required to submit information regarding its litigation history, including litigation with owners and subcontractors.

f) Timeliness

- Firms must provide information on its success at completing projects on time, including the payment of liquidated damages.

g) Capacity

- Firms shall demonstrate sufficient bonding capacity, insurance and resources for the project.
- Firms must provide relevant information on the personnel that will be directly responsible for the work, including the location of the office that will be primarily responsible for work.

h) Safety

- Firms shall also demonstrate an acceptable safety history.
- Firms must provide copies of any complaints, safety violations or reports.

i) Legal Authorization

- The Applicant must provide a copy of License, or provide a statement indicating that they are able to acquire one before submitting a bid on a project. If a statement is required, the Applicant shall identify the states in which they are licensed for this type of work.

Prequalification

Review of Application

a) Prequalification Committee

The Project Manager shall establish a Prequalification Committee for each project to review and score applications submitted by the firms to determine the prequalification eligibility.

b) Review of Applications

- The Prequalification Committee shall use an objective assessment process and criteria for that specific project.
- The prequalification criteria shall not require the firm to have previously been awarded a construction or repair project.
- The prequalification criteria used by the Prequalification Committee shall include prequalification scoring values and the minimum required score to be prequalified for the project.
- The Prequalification Committee shall approve or deny the applications in accordance with the prequalification criteria and scoring system.

c) Notice of Decision

- The firms shall be promptly notified in writing of the Prequalification Committee's decision.
- The notification to a contractor determined not to be prequalified shall include the reason(s) for denial.
- Notice shall be provided prior to the advertisement of bids for the project and with sufficient time for the firm to protest the denial of prequalification.

Prequalification

Selection Process:

A) Quality Based System (QBS)

- The **best and most qualified consultant is selected** as first ranking and negotiation is done for financial term.
- **If negotiation is successful job is awarded to first ranking** if not second ranking consultant is called for negotiation.

B) Quality and Cost Based System (QCBS)

- Evaluation is done considering **both technical capacity and financial terms** in accordance to the weightage given to each categories.
- The consultant with **highest combined score** will be awarded for the job.

C) Cost Based System (CBS)

- Among the technically qualified consultant, job is awarded to the consultant with lowest financial proposal .

Prequalification of Contractor

Following are Considerations for Prequalification of Contractor.

1) Turnover of Contractors:

Average Turn Over (AAT) = $(2 \times V) / T$

Where, V= Value of contract & T= Time in years

If AAT of a contractor is less, Joint venture may be the solution for bidding (PQ).

2) Similar Work Experience: It is taken as $(0.8 V)$. Where, V is the value of contract.

3) Equipment: Types and number of equipment accommodated by the contractor.

4) Financial Status: Net worth must be positive. Sufficient working capital (minimum of 4 to 6 months)

5) Manpower: Once PQ is carried out; list of prequalified contractor is published or announced.

Advantages and Disadvantages of PQ

Advantages:

- To **avoid expense of bidding** for insufficiently qualified contractor/ consultant or to form a joint venture that may give a better chance of success.
- With prequalification, **well qualified firms** will price their bids with the knowledge that they are competing against other qualified bidders meeting realistic minimum competence criteria.
- It helps to **expose potential conflicts of interest**.
- It **reduces the time** to evaluate the bids.
- It **encourages the local firms**
- It **reduces** significantly **problems of rejection** associated to low priced bid.

Disadvantages:

- It may **increase** the **procurement lead time**.
- The employer **is required** to review all PQ applications for **shortlisting**
- The element of **subjective judgment** required by evaluators when applying the prequalification criteria to a number of applicants.

Post Qualification (Post-Q)

- If qualification is done after bidding, it is called post qualification.
- Contractor submit both technical and financial proposal together. First technical qualification (proposal) is evaluated then only the bids of qualified bidders are opened.
- Bids of disqualified bidders will be returned unopened.
- There are **two options**:

1) Single Stage One Envelop

- In this procedure, bidders submit bids in one envelop containing both the financial and technical proposal. The envelopes are opened in public at the specified date and time.
- The financial lowest bidder is identified first and then after evaluation is done in technical aspect whether the lowest bidder is technically qualified.
- If the lowest bidder is not technically qualified, technical proposal of second lowest bidder is evaluated and so on to award the job.

Post Qualification (Post-Q)

2) Single Stage Two Envelop:

- In this procedure, bidders submit two sealed envelopes simultaneously, one financial and other technical proposal, enclosed together in an outer single envelop.
- Initially, technical proposals are opened and evaluated at the specified date and time. The financial proposal remains sealed.
- Financial proposals then opened in public at specified date and time for only the bidders who passed technically. Among them lowest price bid is selected.

Tender Evaluation, Selection and Award of Bidder

Tender Evaluation

- Once bid is opened, that has to be checked against arithmetical corrections, signature of authorized persons, bid security (2 – 3 % of value of contract) and its validity etc. to find out the lowest responsive bid to award the contract.
- Bid security is refunded to unsuccessful bidders. If successful bidder fails to carry out the contract, this amount is lost.

Selection and Award of Bidder

- Once the **lowest responsive bid** is identified and recommended for award of job by bid evaluation committee, decision is taken for award of contract.
- A Letter of Acceptance (**LOA**) is **issued** requesting the selected bidder to submit **performance security** (5 – 13 % value of contract) and to sign the contract.
- Once the **contract is signed** (awarded), **bid security** of other bidders will be **returned**.